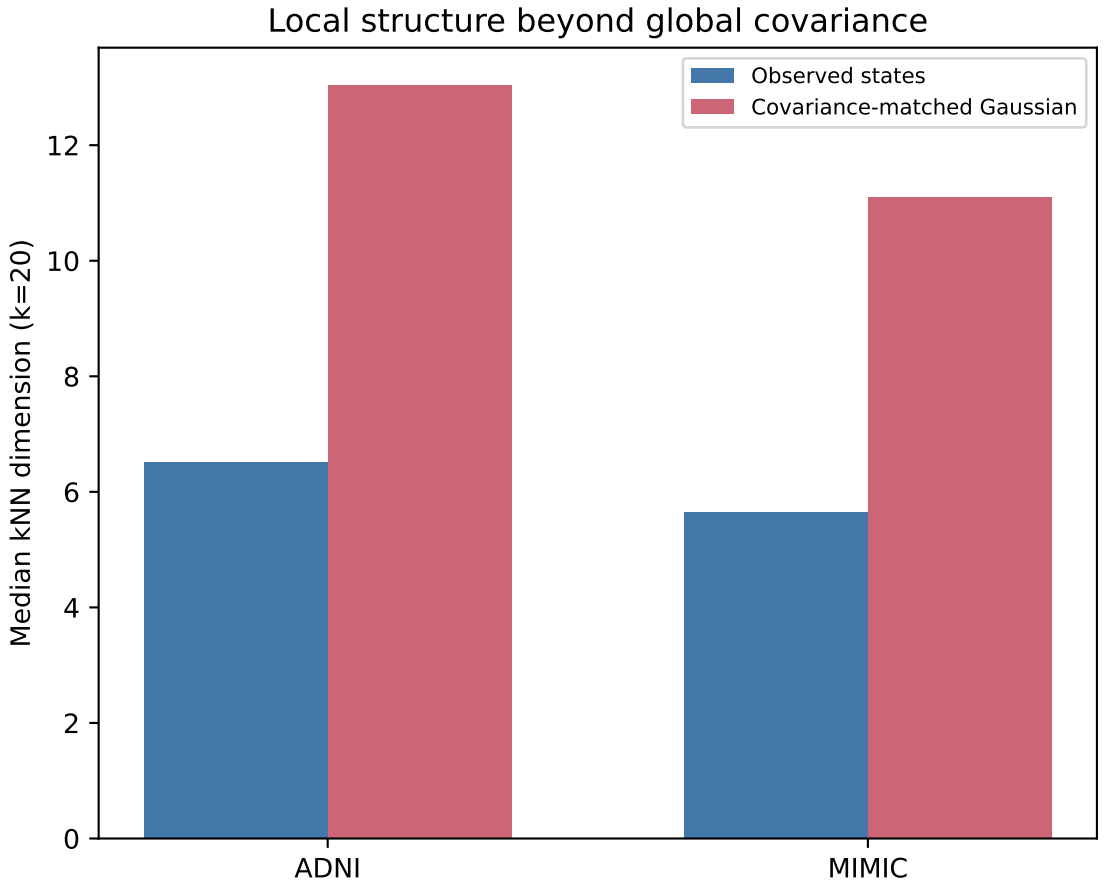
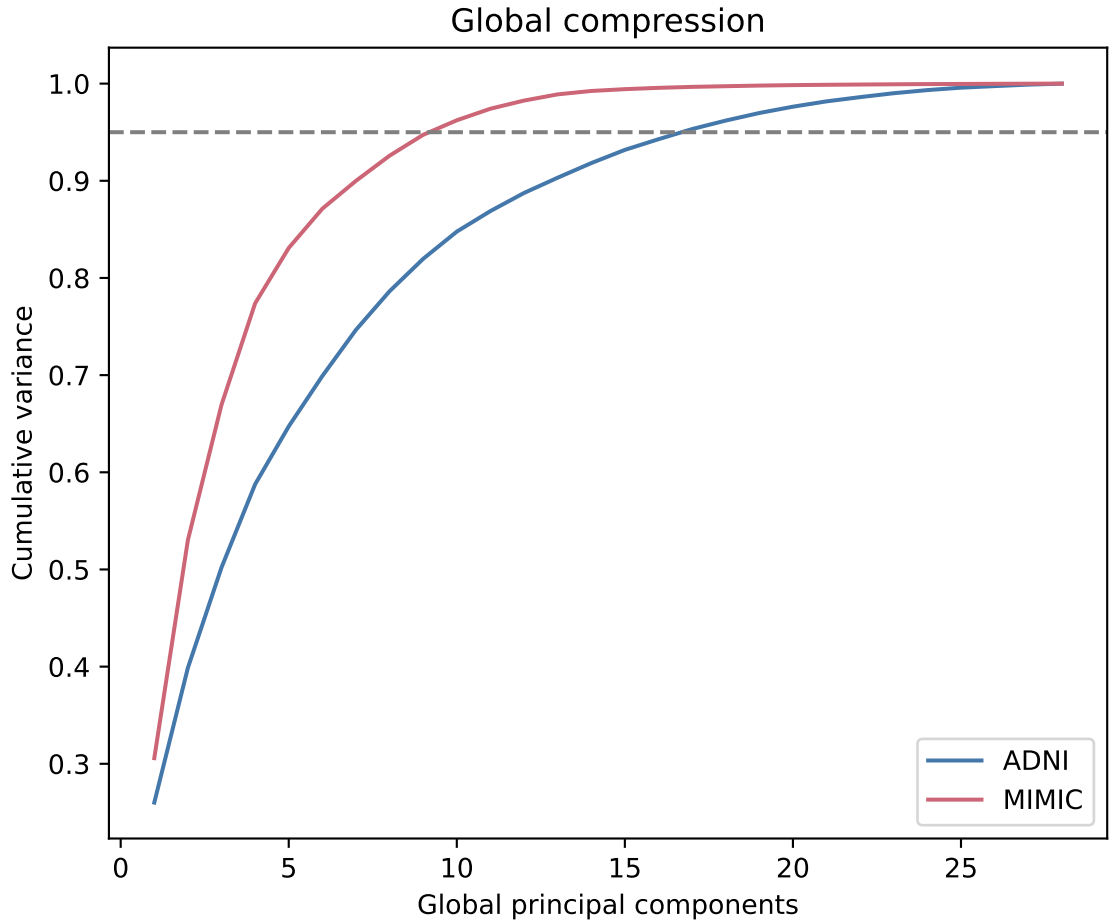
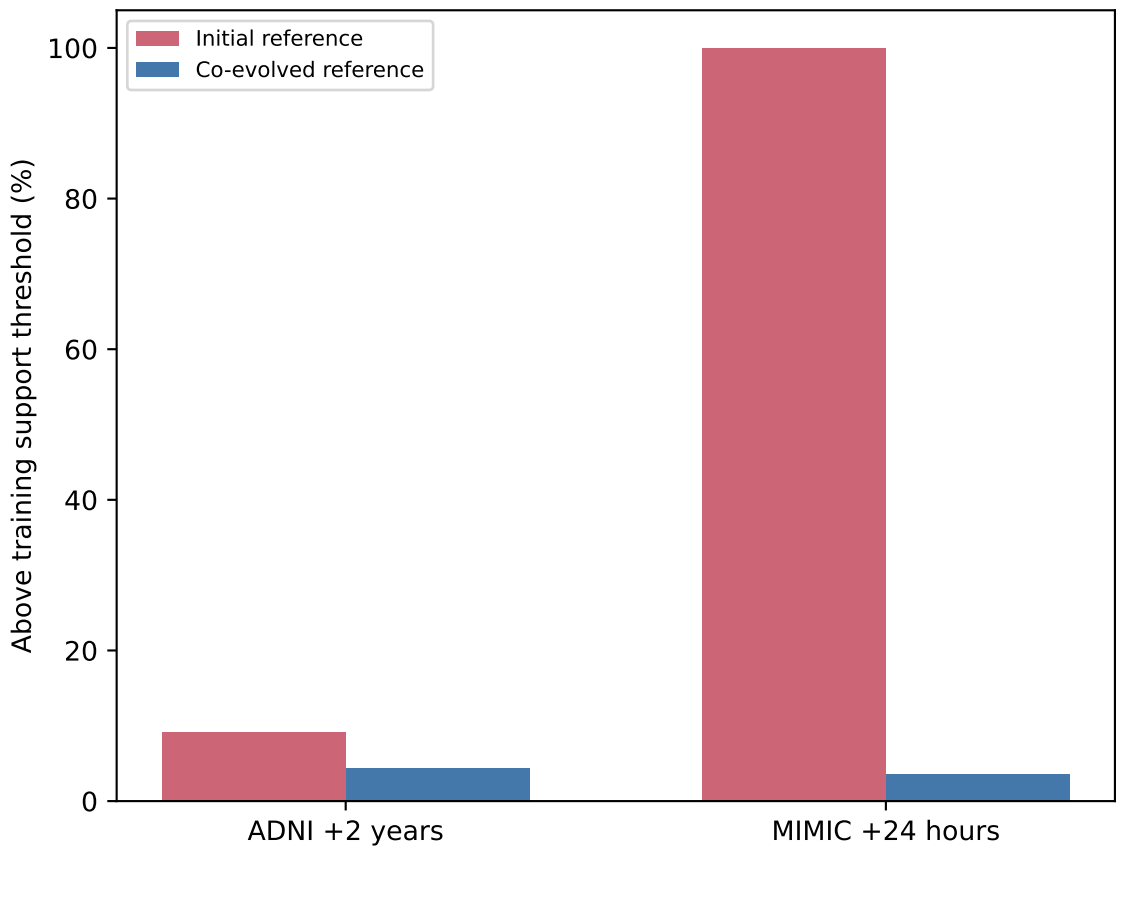
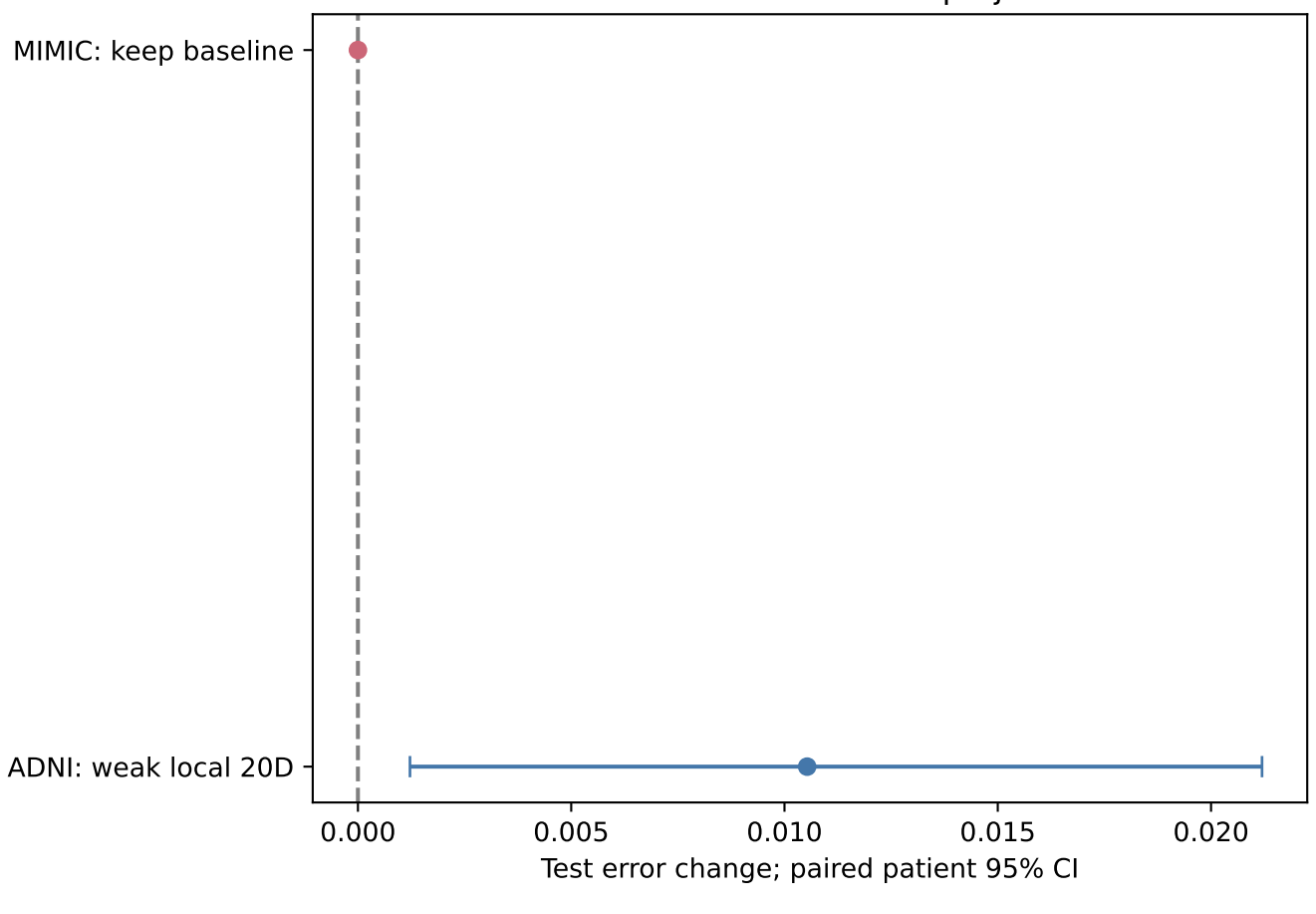

Local structure beyond global covariance
Observed states
Covariance-matched Gaussian
Median kNN dimension (k=20)
0
2
4
6
8
10
12
ADNI
MIMIC


Global compression
Cumulative variance
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
5
10
15
20
25
Global principal components
ADNI
MIMIC


The reference manifold moves with the ODE
Initial reference
Co-evolved reference
Above training support threshold (%)
0
20
40
60
80
100
ADNI +2 years
MIMIC +24 hours


Validation-selected frozen projection
MIMIC: keep baseline
ADNI: weak local 20D
0.000
0.005
0.010
0.015
0.020
Test error change; paired patient 95% CI